
33 Machine-Learning Algorithms

Machine learning may be viewed as a subfield of artificial intelligence. Broadly speaking, artificial intelligence aims to enable computers to carry out complex perception and information-processing tasks with human-like performance. The field of AI is vast and uses many different algorithmic methods.

Machine learning is rich and fascinating, with strong ties to statistics and optimization. Technology today produces enormous amounts of data, providing myriad opportunities for machine-learning algorithms to formulate and test hypotheses about patterns within the data. These hypotheses can then be used to make predictions about the characteristics or classifications in new data. Because machine learning is particularly good with challenging tasks involving uncertainty, where observed data follows unknown rules, it has markedly transformed fields such as medicine, advertising, and speech recognition.

This chapter presents three important machine-learning algorithms: k -means clustering, multiplicative weights, and gradient descent. You can view each of these tasks as a learning problem, whereby an algorithm uses the data collected so far to produce a hypothesis that describes the regularities learned and/or makes predictions about new data. The boundaries of machine learning are imprecise and evolving—some might say that the k -means clustering algorithm should be called “data science” and not “machine learning,” and gradient descent, though an immensely important algorithm for machine learning, also has a multitude of applications outside of machine learning (most notably for optimization problems).

Machine learning typically starts with a *training phase* followed by a *prediction phase* in which predictions are made about new data. For *online learning*, the training and prediction phases are intermingled. The training phase takes as input *training data*, where each input data point has an associated output or *label*; the label might be a category name or some real-valued attribute. It then produces as an output one or more *hypotheses* about how the labels depend on the attributes of the input data points. Hypotheses can take many forms, typically some type of formula or algorithm. The learning algorithm used is often a form of gradient descent. The prediction phase then uses the hypothesis on new data in order to make *predictions* regarding the labels of new data points.

The type of learning just described is known as *supervised learning*, since it starts with a set of inputs that are each labeled. As an example, consider a machine-learning algorithm to recognize spam emails. The training data comprises a collection of emails, each of which is labeled either “spam” or “not spam.” The machine-learning algorithm frames a hypothesis, possibly a rule of the form “if an email has one of a set of words, then it is likely to be spam.” Or it might learn rules that assign a spam score to each word and then evaluates a document by the sum of the spam scores of its constituent words, so that a document with a total score above a certain threshold value is classified as spam. The machine-learning algorithm can then predict whether a new email is spam or not.

A second form of machine learning is *unsupervised learning*, where the training data is unlabeled, as in the clustering problem of Section 33.1. Here the machine-learning algorithm produces hypotheses regarding the centers of groups of input data points.

A third form of machine learning (not covered further here) is *reinforcement learning*, where the machine-learning algorithm takes actions in an environment, receives feedback for those actions from the environment, and then updates its model of the environment based on the feedback. The learner is in an environment that has some state, and the actions of the learner have an effect on that state. Reinforcement

learning is a natural choice for situations such as game playing or operating a self-driving car.

Sometimes the goal in a supervised machine-learning application is not making accurate predictions of labels for new examples, but rather performing causal *inference*: finding an explanatory model that describes how the various features of an input data point affect its associated label. Finding a model that fits a given set of training data well can be tricky. It may involve sophisticated optimization methods that need to balance between producing a hypothesis that fits the data well and producing a hypothesis that is simple.

This chapter focuses on three problem domains: finding hypotheses that group the input data points well (using a clustering algorithm), learning which predictors (experts) to rely upon for making predictions in an online learning problem (using the multiplicative-weights algorithm), and fitting a model to data (using gradient descent).

Section 33.1 considers the clustering problem: how to divide a given set of n training data points into a given number k of groups, or “clusters,” based on a measure of how similar (or more accurately, how dissimilar) points are to each other. The approach is iterative, beginning with an arbitrary initial clustering and incorporating successive improvements until no further improvements occur. Clustering is often used as an initial step when working on a machine-learning problem to discover what structure exists in the data.

Section 33.2 shows how to make online predictions quite accurately when you have a set of predictors, often called “experts,” to rely on, many of which might be poor predictors, but some of which are good predictors. At first, you do not know which predictors are poor and which are good. The goal is to make predictions on new examples that are nearly as good as the predictions made by the best predictor. We study an effective multiplicative-weights prediction method that associates a positive real weight with each predictor and multiplicatively decreases the weights associated with predictors when they make poor predictions. The model in this section is online (see Chapter 27): at each step, we do not know anything about the future examples. In addition, we are able to make predictions even in the presence of adversarial